

M5-CV W3C CREDENTIAL, PRIVACY AND ZERO-KNOWLEDGE REQUIREMENTS MANIFEST

Credential issuer/holder/verifier rules, selective disclosure, notice endpoints and privacy controls

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Human-controlled clean draft • counsel / institutional review as applicable

1. Standards Baseline

M5-CV is designed to express project credentials using W3C-compatible verifiable-credential patterns. W3C Verifiable Credentials Data Model v2.0 is a Recommendation and defines an issuer-holder-verifier model, credential status, evidence, validity, authorization and privacy/security considerations. Conformance to a W3C data format does not by itself make a credential a government-issued credential or confer legal authority.

2. Credential Classes

Credential	Issuer / Evidence	Use
Human principal / M5HUM	Approved identity/authority evidence	Root human authority and delegation.
M5-CV role credential	Employer, agency, licensing body or authorized project issuer	Role/qualification verification.
Government agency issuer credential	Authoritative government identity/PKI or approved trust anchor	Signs agency assertions and M5-DPC notices.
Legal entity credential	Official registry/filing evidence + authorized entity attestation	Entity identity, standing and signatory mapping.
Legal Entity Notice Endpoint	Entity-approved canonical endpoint credential	Receives signed notices/status events where permitted.
Residency / eligibility credential	Approved issuer or evidence workflow	Eligibility proof with selective disclosure / ZK where appropriate.
Professional certification credential	Licensing body / professional issuer	Engineer, architect, environmental, legal or other certification authority.
Project contribution credential	Authorized project/co-op issuer	Work, training, milestone or participation record; not land title.

3. Minimum Credential Fields

- credential ID and type
- issuer ID and authority domain
- subject/holder reference
- claims limited to necessary purpose
- issuance/effective date

- expiration date where applicable
- status/revocation method
- evidence/source reference
- proof/signature
- terms of use / permitted verifier scope where appropriate

4. Privacy and Data Minimization

- Do not publish street address, government ID numbers, financial account data, private signatures or unnecessary PII to a public ledger.
- Use selective disclosure, derived proofs or zero-knowledge techniques only after threat modeling and independent implementation review.
- A ZK proof can reduce disclosure but does not guarantee absolute privacy, anonymity or freedom from compromise.
- Keep revocable links between public proof and private evidence in the controlled M5POD/evidence layer where lawful and necessary.
- Public status should reveal the minimum fact needed: e.g., ELIGIBLE / NOT ELIGIBLE rather than the underlying private record.

5. Government Notice Endpoint Credential

Field	Requirement
entity_tcid	Stable project/legal-entity identifier.
jurisdiction	Applicable entity/notice jurisdiction.
canonical_endpoint	Authenticated endpoint/resolver.
authorized_agent	Current person/role credential authorized to administer endpoint.
notice_classes_accepted	Classes the entity elects/legally may receive electronically.
formal_service_classes	Only classes legally recognized for digital service; otherwise NONE / fallback.
fallback_address_or_agent	Statutory/contractual fallback channel.
effective / expiry	Validity window.
issuer / revocation	Authorized issuer and revocation/status method.

6. Credential Failure Rule

Expired, revoked, out-of-domain or unverifiable credentials fail closed for material actions. M5Canon must route to renewal/cure or authorized human verification rather than infer authority.

Selected Authoritative References

- W3C Verifiable Credentials Data Model v2.0 — Recommendation, May 15, 2025.
- W3C DID Core / related identity standards as applicable to implementation.
- California Secretary of State — official entity records and service-of-process guidance.